

Algebraic tunneling for the restricted fractional Laplacian on distant domains

LESLIE P. POLZER

Independent Researcher

polzer@fastmail.com

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Abstract. We study the restricted fractional Laplacian $(-\Delta)^s$, $0 < s < 1$, on distant translates of a bounded open set $D \subset \mathbb{R}^d$. Let μ_1 be the one-well ground-state eigenvalue, let ϕ_1 be its nonnegative L^2 -normalized eigenfunction, and set $m_1 = \int_D \phi_1$. For $\Omega_L = (D - Le/2) \cup (D + Le/2)$ with a fixed unit vector e , we prove, as $L \rightarrow \infty$,

$$\lambda_{1,2}(\Omega_L) = \mu_1 \mp c_{d,s} m_1^2 L^{-d-2s} + O(L^{-d-2s-2} + L^{-2d-4s}),$$

where the minus sign corresponds to λ_1 and $c_{d,s}$ is the standard singular-integral normalization. We also give an explicit separation threshold beyond which both eigenvalues are simple and satisfy explicit remainder bounds, and, when $D = -D$, an exact decomposition into symmetric and antisymmetric sectors under central inversion. For fixed $N \geq 2$ and distinct sites $a_1, \dots, a_N$, let $\Omega_{\mathbf{a},L} = \bigcup_{j=1}^N (D + La_j)$. Its first N eigenvalues satisfy $\lambda_k(\Omega_{\mathbf{a},L}) = \mu_1 + L^{-d-2s} \theta_k + O(L^{-d-2s-2} + L^{-2d-4s})$, where $\theta_1 \leq \dots \leq \theta_N$ are the eigenvalues of the matrix with zero diagonal and off-diagonal entries $-c_{d,s} m_1^2 |a_i - a_j|^{-d-2s}$. The estimates follow from spectral-gap bounds and Taylor expansion with a quadratic remainder. For two equal balls, the expansion for λ_2 gives the leading rate in the nonlocal Hong–Kahn–Szegő minimizing sequence. Galerkin computations using exact cell integrals illustrate the two- and three-well asymptotics at fixed mesh. The theorems, the corollaries, and the cell-integral proposition, together with the one-well spectral facts they use, are formally verified in Lean 4 with Mathlib.

Keywords. restricted fractional Laplacian; eigenvalue splitting; nonlocal double wells; spectral perturbation; shape optimization.

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1. Introduction

The restricted fractional Laplacian couples disjoint components through its integral kernel, unlike the local Dirichlet Laplacian, whose spectrum on a disconnected union is exactly the multiset union of the component spectra. Here tunneling refers to the splitting of the one-well ground-state level caused by the nonlocal interaction between distant copies of a reference well. We consider identically oriented translates of a common well; the splitting decays algebraically with their separation.

Theorem 3.3 gives explicit bounds for the two eigenvalue shifts in the centrally symmetric case. Theorem 4.6 approximates the first N eigenvalues for a fixed family of translates by the spectrum of an explicit interaction matrix, with error bounds uniform over those N eigenvalues. Its two-well specialization, Corollary 4.7, gives the doublet asymptotic without

central symmetry, and Corollary 4.8 shows that every conclusion of Theorem 3.3, with the same explicit threshold and remainders, holds for an arbitrary bounded open well. For two equal balls, Corollary 3.4 identifies the leading rate along the Hong–Krahn–Szegő minimizing sequence. Section 5 illustrates the two- and three-well asymptotics by Galerkin computations at fixed mesh.

Central symmetry permits an exact decomposition into the symmetric and antisymmetric sector operators $A_D \pm B_L$. For the general family of translates, the linear term in the kernel expansion cancels after projection onto the one-well ground states. This gives the same remainder order without symmetry. The proofs use the one-well spectral facts recalled in Section 2, together with elementary perturbation estimates. No boundary regularity of the well is needed: Lemma 2.1 gives the compactness of the form embedding for every bounded open set. Section 6 describes a formal verification of the theorems, the corollaries, and Proposition 5.1 in Lean 4 with Mathlib.

Brasco and Parini proved that the fractional Hong–Krahn–Szegő bound for the second eigenvalue is strict and that two equal balls form a minimizing sequence when their mutual distance tends to infinity [BP16, Theorem 6.2]. Goel and Sreenadh obtained an analogue for a local p -Laplacian coupled to a nonlocal p -Laplacian with a compactly supported kernel [GS19, Theorem 1.4]; the latter interaction is finite-range rather than algebraic. Biagi, Dipierro, Valdinoci, and Vecchi proved a mixed local/nonlocal analogue [BDVV23, Theorem 1.1]. Parini and Salort proved a fractional concentration–compactness trichotomy for bounded Sobolev sequences and transferred its dichotomy to quasi-open domains through operator-norm resolvent approximation [PS20, Theorems 1.1 and 1.2]. For two one-dimensional patches, Léculier and Roquejoffre proved monotonicity and continuity of the principal eigenvalue throughout the separation range, including contact [LR23, Theorem 1]. These results describe the qualitative dependence on separation and the limiting shape-optimization behavior.

The influence of mutual position on the restricted fractional spectrum is also emphasized by Abatangelo, Felli, and Noris [AFN20, arXiv v2, Introduction, p. 3], in a study of small fractional-capacity perturbations. In the shape-optimization setting, Zahl isolated the exact cross-energy under translations and proposed a signed point-mass interaction with kernel $|x - y|^{-d-2s}$ [Zah26, Section 10, equations (10.3) and (10.7), and Conjecture 10.9]. Wu subsequently proved that this discrete signed-mass model has no local minima, settling that conjecture [Wu26, Introduction]. Dipierro, Proietti Lippi, Sportelli, and Valdinoci studied disconnected domains for generalized eigenproblems having a superposition of lower-order operators on the right-hand side [DPLSV26, Theorems 1.4–1.7].

Zahl’s discrete energy anticipates the matrix in (4.9). Here we derive that matrix from the standard restricted fractional Laplacian, controlling the interaction in operator norm and the contribution of complementary modes through the one-well spectral gap. We are not aware of an earlier result giving either the two-component coefficient with a quantified remainder or this multi-well spectral reduction.

2. The one-well operator and exact two-well reduction

Fix $d \geq 1$ and $0 < s < 1$, and put

$$\kappa = d + 2s.$$

For an open set $G \subset \mathbb{R}^d$, let

$$\tilde{H}^s(G) = \{u \in H^s(\mathbb{R}^d) : u = 0 \text{ a.e. on } \mathbb{R}^d \setminus G\}.$$

We use the Gagliardo seminorm

$$[u]_{H^s(\mathbb{R}^d)}^2 = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \frac{|u(x) - u(y)|^2}{|x - y|^{d+2s}} dx dy.$$

For bounded Lipschitz G , this space agrees with the completion of $C_c^\infty(G)$ in the global $H^s(\mathbb{R}^d)$ norm [DFW21, Section 2, equation (2.1)]. The zero-exterior quadratic form is

$$\mathcal{Q}_G[u] = \frac{c_{d,s}}{2} \iint_{\mathbb{R}^d \times \mathbb{R}^d} \frac{|u(x) - u(y)|^2}{|x - y|^\kappa} dx dy, \quad u \in \tilde{H}^s(G), \quad (2.1)$$

where $c_{d,s} = 4^s s \Gamma(d/2 + s) / (\pi^{d/2} \Gamma(1 - s))$ is the standard normalization in the singular-integral definition of $(-\Delta)^s$. Brasco and Parini use coefficient 2 in place of $c_{d,s}$ in the singular-integral operator, so their eigenvalues are multiplied here by $c_{d,s}/2$; the positivity, simplicity, ordering, and shape-optimization statements cited below are unchanged by this fixed positive rescaling [BP16, arXiv v2, Section 1.1, p. 2]. The form norm $(\mathcal{Q}_G[u] + \|u\|_{L^2(G)}^2)^{1/2}$ is equivalent to the norm inherited from $H^s(\mathbb{R}^d)$ on $\tilde{H}^s(G)$. Thus (2.1) is a densely defined closed form, and we denote its associated self-adjoint operator on $L^2(G)$ by A_G . We write $\lambda_1(G) \leq \lambda_2(G) \leq \dots$ for the eigenvalues of A_G , repeated according to multiplicity. For an abstract self-adjoint operator T with compact resolvent, $\lambda_k(T)$ has the analogous meaning.

For Lipschitz G the compactness of the form embedding follows from the extension property and [DNPV12, Theorems 5.4 and 7.1]. The following cube-averaging argument needs no boundary regularity.

Lemma 2.1 (*Compactness of the form embedding*). *Let $G \subset B_R(0)$ be open. Every sequence in $\tilde{H}^s(G)$ that is bounded in the form norm has a subsequence converging in $L^2(\mathbb{R}^d)$. In particular, A_G has compact resolvent.*

Proof. Fix $\varepsilon > 0$, tile $\mathbb{R}^d$ by the half-open cubes Q of side ε with vertices in $\varepsilon\mathbb{Z}^d$, and put $P_\varepsilon u = \sum_Q (\varepsilon^{-d} \int_Q u) \mathbf{1}_Q$. Expanding the squares gives, for each cube,

$$\begin{aligned} \int_Q \left| u - \varepsilon^{-d} \int_Q u \right|^2 &= \frac{1}{2\varepsilon^d} \iint_{Q \times Q} |u(x) - u(y)|^2 dx dy \\ &\leq \frac{(\sqrt{d}\varepsilon)^\kappa}{2\varepsilon^d} \iint_{Q \times Q} \frac{|u(x) - u(y)|^2}{|x - y|^\kappa} dx dy, \end{aligned}$$

because $|x - y| \leq \sqrt{d}\varepsilon$ on $Q \times Q$. The sets $Q \times Q$ are pairwise disjoint, so summation over the cubes gives

$$\|u - P_\varepsilon u\|_{L^2(\mathbb{R}^d)}^2 \leq d^{\kappa/2} \varepsilon^{2s} \frac{\mathcal{Q}_G[u]}{c_{d,s}}, \quad u \in \tilde{H}^s(G).$$

Only the finitely many cubes meeting $B_R(0)$ contribute to $P_\varepsilon u$, and $\|P_\varepsilon u\|_{L^2} \leq \|u\|_{L^2}$ by the Cauchy–Schwarz inequality. Hence a set bounded in the form norm lies within distance $C\varepsilon^s$ of a bounded subset of a finite-dimensional space. Since $\varepsilon > 0$ is arbitrary, the set is totally bounded, hence relatively compact in the complete space $L^2(\mathbb{R}^d)$. Compactness of the form embedding gives the compactness of the resolvent of A_G . ■

The following facts are proved directly for the space $\tilde{H}^s(G)$ above.

Lemma 2.2 (*Ground state*). *Let $G \subset B_R(0)$ be a nonempty open set. Then $\lambda_1(G) < \lambda_2(G)$, every eigenfunction of $\lambda_1(G)$ is either nonnegative or nonpositive almost everywhere, and the eigenspace of $\lambda_1(G)$ is spanned by a normalized eigenfunction $\phi \geq 0$ with $\int_G \phi > 0$.*

Proof. By Lemma 2.1, A_G has compact resolvent, and the eigenfunctions of $\lambda_1(G)$ are exactly the nonzero minimizers of the Rayleigh quotient $\mathcal{Q}_G[u]/\|u\|_{L^2}^2$ on $\tilde{H}^s(G)$. For real u and all x, y ,

$$|u(x) - u(y)|^2 - (|u(x)| - |u(y)|)^2 = 2(|u(x)u(y)| - u(x)u(y)) \geq 0.$$

Hence $|u| \in \tilde{H}^s(G)$ and $\mathcal{Q}_G[|u|] \leq \mathcal{Q}_G[u]$, with equality only if $u(x)u(y) \geq 0$ for almost every (x, y) , because the kernel is positive off the diagonal. By Tonelli's theorem, the product of the measures of $\{u > 0\}$ and $\{u < 0\}$ then vanishes, so u has constant sign. If u is an eigenfunction of $\lambda_1(G)$, then $|u|$ is also a minimizer, so equality holds and u has constant sign. If the eigenspace contained two orthonormal eigenfunctions, we could choose them nonnegative; their orthogonality would force disjoint supports up to null sets, and their difference would be an eigenfunction that changes sign, a contradiction. Thus $\lambda_1(G)$ is simple. A normalized nonnegative eigenfunction is nonzero and integrable on the bounded set G , so its integral is positive. ■

Let $D \subset B_R(0)$ be a nonempty bounded open set. By Lemmas 2.1 and 2.2, A_D has compact resolvent and simple first eigenvalue. Put

$$\mu_1 = \lambda_1(D), \quad \mu_2 = \lambda_2(D), \quad g = \mu_2 - \mu_1 > 0.$$

Fix the nonnegative normalized ground-state eigenfunction and set

$$A_D \phi_1 = \mu_1 \phi_1, \quad \|\phi_1\|_{L^2(D)} = 1, \quad m_1 = \int_D \phi_1(x) dx > 0. \quad (2.2)$$

For Lipschitz D , the space $\tilde{H}^s(D)$ is the completion used in [BP16] by [DFW21, Section 2, equation (2.1)], and [BP16, Theorem 2.8], specialized to the linear case, shows that $\phi_1 > 0$ almost everywhere in D . The arguments below use only $\phi_1 \geq 0$. When $D = -D$, simplicity

and invariance under central inversion show that $\phi_1(-x) = \pm\phi_1(x)$; since ϕ_1 is nonnegative and nonzero, $\phi_1(-x) = \phi_1(x)$.

Fix a unit vector $e \in \mathbb{R}^d$. For $L > 2R$, define

$$D_L^- = D - \frac{L}{2}e, \quad D_L^+ = D + \frac{L}{2}e, \quad \Omega_L = D_L^- \cup D_L^+.$$

On $L^2(D)$ define the bounded integral operator

$$(B_L v)(x) = -c_{d,s} \int_D \frac{v(y)}{|Le - x - y|^\kappa} dy. \quad (2.3)$$

Its kernel is real and symmetric, so B_L is self-adjoint.

Lemma 2.3 (*Exact reduction by central inversion*). *Assume $D = -D$. The operator A_{Ω_L} is unitarily equivalent to*

$$\begin{pmatrix} A_D & B_L \\ B_L & A_D \end{pmatrix} \quad \text{on } L^2(D) \oplus L^2(D),$$

and hence to

$$(A_D + B_L) \oplus (A_D - B_L). \quad (2.4)$$

Proof. For $f \in \tilde{H}^s(\Omega_L)$ set

$$u(x) = f\left(x - \frac{L}{2}e\right), \quad v(x) = f\left(-x + \frac{L}{2}e\right), \quad x \in D.$$

This defines a unitary map from $L^2(\Omega_L)$ to $L^2(D) \oplus L^2(D)$. Translation invariance and invariance under central inversion identify the two diagonal parts of the form with $\mathcal{Q}_D[u]$ and $\mathcal{Q}_D[v]$.

To identify the form domains, fix $\chi \in C_c^\infty(\mathbb{R}^d)$ and $w \in H^s(\mathbb{R}^d)$. Then

$$\begin{aligned} [\chi w]_{H^s(\mathbb{R}^d)}^2 &\leq 2\|\chi\|_{L^\infty}^2 [w]_{H^s(\mathbb{R}^d)}^2 \\ &\quad + 2 \int_{\mathbb{R}^d} |w(y)|^2 \int_{\mathbb{R}^d} \frac{|\chi(x) - \chi(y)|^2}{|x - y|^{d+2s}} dx dy. \end{aligned}$$

The inner integral is bounded uniformly in y : use $|\chi(x) - \chi(y)| \leq \|\nabla\chi\|_{L^\infty}|x - y|$ when $|x - y| \leq 1$ and $|\chi(x) - \chi(y)| \leq 2\|\chi\|_{L^\infty}$ otherwise. The two radial bounds are integrable because $s < 1$ and $s > 0$, respectively. Consequently multiplication by χ is bounded on $H^s(\mathbb{R}^d)$.

Positive separation permits cutoffs $\chi_\pm$ that equal one on $D_L^\pm$ and vanish near the other component. For $f \in \tilde{H}^s(\Omega_L)$, the functions $\chi_- f$ and $\chi_+ f$ therefore belong to $H^s(\mathbb{R}^d)$ and, after the above changes of variables, give $u, v \in \tilde{H}^s(D)$. Conversely, translated and reflected zero extensions of any $u, v \in \tilde{H}^s(D)$ belong to $H^s(\mathbb{R}^d)$, and their sum is supported in Ω_L . Hence the unitary identifies $\tilde{H}^s(\Omega_L)$ with $\tilde{H}^s(D) \oplus \tilde{H}^s(D)$.

The cross-kernel is bounded because of positive separation. Each separate zero-exterior form already includes the squared-modulus terms integrated against the kernel over the

other component. Thus, after the above changes of variables, expanding the square in (2.1) yields the exact identity

$$\begin{aligned} & \mathcal{Q}_{\Omega_L}[f] - \mathcal{Q}_D[u] - \mathcal{Q}_D[v] \\ &= c_{d,s} \iint_{D \times D} \frac{|u(x) - v(y)|^2 - |u(x)|^2 - |v(y)|^2}{|Le - x - y|^\kappa} dx dy \\ &= -2c_{d,s} \operatorname{Re} \iint_{D \times D} \frac{u(x) \overline{v(y)}}{|Le - x - y|^\kappa} dx dy = 2 \operatorname{Re} \langle u, B_L v \rangle. \end{aligned}$$

The square terms therefore cancel, leaving no additional diagonal contribution. This proves the block representation in the sense of closed forms. Since B_L is bounded, it also gives the stated operator representation. The unitary change of variables

$$(u, v) \mapsto 2^{-1/2}(u + v, u - v)$$

yields (2.4). ■

3. The doublet splitting

We first record the elementary perturbation estimate used for the two branches.

Lemma 3.1 (*Ground-state perturbation bound*). *Let A be self-adjoint with compact resolvent, simple lowest eigenvalue μ , and normalized ground state ϕ . Define the spectral gap by $g = \lambda_2(A) - \mu > 0$. Let W be bounded and self-adjoint, put $b = \|W\|$ and $w = \langle \phi, W\phi \rangle$, and suppose $b \leq g/4$. Then*

$$-\frac{2b^2}{g} \leq \lambda_1(A + W) - (\mu + w) \leq 0. \quad (3.1)$$

Proof. Let $\mathfrak{a}$ be the closed quadratic form associated with A . The upper bound follows by testing the Rayleigh quotient with ϕ . For a normalized vector $\psi = a\phi + \eta$ in the form domain of $\mathfrak{a}$, where $\eta \perp \phi$ and $t = \|\eta\|$, the spectral gap gives

$$\mathfrak{a}[\psi] \geq \mu + gt^2.$$

Since $|a|^2 = 1 - t^2$, self-adjointness gives

$$\langle \psi, W\psi \rangle - w = -t^2 w + 2 \operatorname{Re} \langle a\phi, W\eta \rangle + \langle \eta, W\eta \rangle.$$

Using $|w| \leq b$, $|a| \leq 1$, and $\|W\| = b$,

$$\begin{aligned} \mathfrak{a}[\psi] + \langle \psi, W\psi \rangle - (\mu + w) &\geq gt^2 - 2bt^2 - 2bt \\ &= (g - 2b)t^2 - 2bt \\ &= (g - 2b) \left(t - \frac{b}{g - 2b} \right)^2 - \frac{b^2}{g - 2b} \\ &\geq -\frac{b^2}{g - 2b} \geq -\frac{2b^2}{g}. \end{aligned}$$

Taking the infimum over normalized ψ proves the lower bound. ■

The next lemma extracts the interaction coefficient. In the reflected coordinates used here, evenness of the ground state removes the linear Taylor term. Section 4 obtains the same error order without symmetry from an algebraic cancellation in translated coordinates.

Lemma 3.2 (*Interaction-kernel expansion*). Assume $D = -D$, and let

$$I_L = \iint_{D \times D} \frac{\phi_1(x)\phi_1(y)}{|Le - x - y|^\kappa} dx dy.$$

For $L \geq 4R$,

$$|I_L - m_1^2 L^{-\kappa}| \leq C_{\kappa,R} m_1^2 L^{-\kappa-2}, \quad C_{\kappa,R} = 2^{\kappa+3} \kappa(\kappa+1) R^2. \quad (3.2)$$

Moreover,

$$\|B_L\| \leq \beta_L, \quad \beta_L = c_{d,s} 2^\kappa |D| L^{-\kappa}. \quad (3.3)$$

Proof. Set $F_L(z) = |Le - z|^{-\kappa}$. For $|z| \leq 2R$ and $L \geq 4R$, the segment from 0 to z remains at distance at least $L/2$ from Le . Put $n_z = (Le - z)/|Le - z|$. Direct differentiation gives

$$\nabla^2 F_L(z) = \kappa |Le - z|^{-\kappa-2} ((\kappa+2)n_z \otimes n_z - I).$$

Its radial and transverse eigenvalues are respectively $\kappa(\kappa+1)|Le - z|^{-\kappa-2}$ and $-\kappa|Le - z|^{-\kappa-2}$. Since $\kappa > 1$,

$$\|\nabla^2 F_L(z)\| = \kappa(\kappa+1)|Le - z|^{-\kappa-2}.$$

Taylor's theorem therefore yields

$$|F_L(z) - L^{-\kappa} - \kappa L^{-\kappa-1} e \cdot z| \leq C_{\kappa,R} L^{-\kappa-2}$$

for $|z| \leq 2R$, with the displayed value of $C_{\kappa,R}$. Because ϕ_1 is even,

$$\int_D x \phi_1(x) dx = 0.$$

Consequently the integral of the linear Taylor term with $z = x + y$ vanishes. Since ϕ_1 is nonnegative, integration of the remainder gives (3.2).

Finally, $|Le - x - y| \geq L - 2R \geq L/2$. The Schur test applied to (2.3) gives

$$\|B_L\| \leq c_{d,s} |D| (L - 2R)^{-\kappa} \leq c_{d,s} 2^\kappa |D| L^{-\kappa},$$

which is (3.3). ■

Theorem 3.3 (*Two-well algebraic splitting*). Let $D \subset B_R(0)$ be a nonempty bounded centrally symmetric open set, and let Ω_L be as in Section 2. Set

$$L_0 = \max \left\{ 4R, \left(\frac{4c_{d,s} 2^\kappa |D|}{g} \right)^{1/\kappa}, 2 \left(\frac{2c_{d,s} 4^\kappa |D|^2}{gm_1^2} \right)^{1/\kappa} \right\}. \quad (3.4)$$

For $L \geq L_0$, the first two eigenvalues of A_{Ω_L} satisfy

$$|\lambda_1(\Omega_L) - (\mu_1 - c_{d,s}m_1^2L^{-\kappa})| \leq c_{d,s}C_{\kappa,R}m_1^2L^{-\kappa-2} + \frac{2\beta_L^2}{g}, \quad (3.5)$$

$$|\lambda_2(\Omega_L) - (\mu_1 + c_{d,s}m_1^2L^{-\kappa})| \leq c_{d,s}C_{\kappa,R}m_1^2L^{-\kappa-2} + \frac{2\beta_L^2}{g}. \quad (3.6)$$

Both eigenvalues are simple, and

$$\lambda_3(\Omega_L) \geq \mu_1 + \frac{3g}{4}, \quad \lambda_3(\Omega_L) - \lambda_2(\Omega_L) \geq \frac{g}{2}. \quad (3.7)$$

In particular, as $L \rightarrow \infty$,

$$L^{d+2s}(\lambda_2(\Omega_L) - \lambda_1(\Omega_L)) \longrightarrow 2c_{d,s}m_1^2. \quad (3.8)$$

Proof. By Lemma 2.3, the spectrum is the union of the spectra of $A_D + B_L$ and $A_D - B_L$. Put

$$t_L = \langle \phi_1, B_L \phi_1 \rangle = -c_{d,s}I_L < 0.$$

For $L \geq L_0$, the second term in (3.4) gives the first inequality in

$$\beta_L \leq \frac{g}{4}, \quad 2c_{d,s}m_1^2(L + 2R)^{-\kappa} > \frac{4\beta_L^2}{g}. \quad (3.9)$$

Indeed, $L \geq 4R$ gives $L + 2R \leq 3L/2$, while the third term in (3.4) and $3/2 < 2$ give

$$L^\kappa > \left(\frac{3}{2}\right)^\kappa \frac{2c_{d,s}4^\kappa|D|^2}{gm_1^2}.$$

Substitution of $\beta_L = c_{d,s}2^\kappa|D|L^{-\kappa}$ proves the strict second inequality in (3.9).

Apply Lemma 3.1 first with $W = B_L$ and then with $W = -B_L$. Let $E_+(L)$ and $E_-(L)$ denote the ground-state energies of the symmetric and antisymmetric branches, respectively. They have the form

$$E_+(L) = \mu_1 + t_L + \rho_+(L), \quad E_-(L) = \mu_1 - t_L + \rho_-(L), \quad (3.10)$$

where

$$|\rho_\pm(L)| \leq \frac{2\beta_L^2}{g}.$$

The second eigenvalue of either branch is at least $\mu_2 - \beta_L \geq \mu_1 + 3g/4$, whereas both branch ground-state energies are at most $\mu_1 + \beta_L \leq \mu_1 + g/4$. Thus the first two eigenvalues of the full operator are the two numbers in (3.10).

Since $|Le - x - y| \leq L + 2R$ and $\phi_1 \geq 0$,

$$I_L \geq m_1^2(L + 2R)^{-\kappa}.$$

The second condition in (3.9) and (3.10) imply $E_+(L) < E_-(L)$. Hence $\lambda_1(\Omega_L) = E_+(L)$ and $\lambda_2(\Omega_L) = E_-(L)$. The within-branch separation above makes both eigenvalues simple. It also gives $\lambda_3(\Omega_L) \geq \mu_1 + 3g/4$, while $\lambda_2(\Omega_L) \leq \mu_1 + g/4$; this proves (3.7). Equations (3.5) and (3.6) now follow from Lemma 3.2. Subtracting the two expansions and multiplying by L^κ proves (3.8), since $\kappa > 0$. $\blacksquare$

Central symmetry enters the proof only through Lemmas 2.3 and 3.2. Corollary 4.8 proves every conclusion of Theorem 3.3, with the same threshold L_0 and the same remainders, for an arbitrary nonempty bounded open set $D \subset B_R(0)$, by the translated-coordinate argument of Section 4.

Corollary 3.4 (*Rate for the distant-ball minimizing sequence*). *Let $D = B_R(0)$ and let Ω_L be the union of two radius- R balls whose centers are separated by L , so that $|\Omega_L| = 2|B_R|$. Then, as $L \rightarrow \infty$,*

$$L^{d+2s}(\lambda_2(\Omega_L) - \lambda_1(B_R)) \rightarrow c_{d,s} \left(\int_{B_R} \phi_1(x) dx \right)^2.$$

Thus the minimizing sequence in [BP16, Theorem 6.2] approaches its limiting value at the explicit leading order L^{-d-2s} .

Proof. This is (3.6) with $D = B_R(0)$ and $\mu_1 = \lambda_1(B_R)$. ■

4. Finitely many wells

The two-well decomposition induced by central inversion is specific to a pair. For finitely many components we instead keep the full block operator and compare its lowest spectral cluster with its compression to the translated one-well ground states.

Throughout this section, let $D \subset B_R(0)$ be any nonempty bounded open set; central symmetry is not assumed. Lemmas 2.1 and 2.2 apply unchanged. Write $\mu_1 < \mu_2$, $g = \mu_2 - \mu_1 > 0$, choose the nonnegative normalized ground state $A_D \phi_1 = \mu_1 \phi_1$, and set $m_1 = \int_D \phi_1(x) dx > 0$. Every $L \rightarrow \infty$ statement in this section keeps N and the distinct sites $\mathbf{a}$ fixed.

Definition 4.1 (Multi-well geometry). Fix an integer $N \geq 2$ and distinct points $\mathbf{a} = (a_1, \dots, a_N)$ in $\mathbb{R}^d$. Set

$$\delta_{\mathbf{a}} = \min_{i \neq j} |a_i - a_j|, \quad \Sigma_p(\mathbf{a}) = \max_{1 \leq i \leq N} \sum_{j \neq i} |a_i - a_j|^{-p} \quad (p > 0). \quad (4.1)$$

For $L\delta_{\mathbf{a}} > 2R$, define

$$D_{j,L} = D + La_j, \quad \Omega_{\mathbf{a},L} = \bigcup_{j=1}^N D_{j,L}. \quad (4.2)$$

The sets in this union are pairwise disjoint.

On $\mathcal{H}_N = \bigoplus_{j=1}^N L^2(D)$, let $\mathcal{A}_0 = \bigoplus_{j=1}^N A_D$. For $i \neq j$ define

$$(B_{ij,L}v)(x) = -c_{d,s} \int_D \frac{v(y)}{|L(a_i - a_j) + x - y|^\kappa} dy, \quad B_{ii,L} = 0, \quad (4.3)$$

and let $\mathcal{V}_L = (B_{ij,L})_{i,j=1}^N$.

Lemma 4.2 (*Exact multi-well reduction and norm bound*). *If $L\delta_{\mathbf{a}} \geq 4R$, then $A_{\Omega_{\mathbf{a},L}}$ is unitarily equivalent to $\mathcal{A}_0 + \mathcal{V}_L$. Moreover,*

$$\|\mathcal{V}_L\| \leq \gamma_L, \quad \gamma_L = c_{d,s} 2^\kappa |D| \Sigma_\kappa(\mathbf{a}) L^{-\kappa}. \quad (4.4)$$

Proof. For $f \in L^2(\Omega_{\mathbf{a},L})$, put $u_j(x) = f(x + La_j)$ for $x \in D$. This is a unitary identification with $\mathcal{H}_N$. Translation invariance gives the diagonal operator $\mathcal{A}_0$. The cutoff argument in the proof of Lemma 2.3 identifies the form domain with $\bigoplus_{j=1}^N \tilde{H}^s(D)$, and positive separation makes every cross-kernel bounded. For $i < j$, direct expansion of the form (2.1) gives the cross term

$$-2c_{d,s} \operatorname{Re} \iint_{D \times D} \frac{u_i(x) \overline{u_j(y)}}{|L(a_i - a_j) + x - y|^\kappa} dx dy.$$

Since $B_{ji,L} = B_{ij,L}^*$, summing these terms proves the exact block representation in the sense of closed forms and hence as operators.

For $i \neq j$ and $x, y \in D$,

$$|L(a_i - a_j) + x - y| \geq L|a_i - a_j| - 2R \geq \frac{L}{2}|a_i - a_j|.$$

The Schur test therefore gives

$$\|B_{ij,L}\| \leq c_{d,s} 2^\kappa |D| L^{-\kappa} |a_i - a_j|^{-\kappa}. \quad (4.5)$$

For $u = (u_1, \dots, u_N)$, set $r_j = \|u_j\|_{L^2(D)}$ and let $\mathbf{N}_L$ be the symmetric matrix with entries $(\mathbf{N}_L)_{ij} = \|B_{ij,L}\|$. Then

$$\|\mathcal{V}_L u\|_{\mathcal{H}_N} \leq \|\mathbf{N}_L r\|_{\ell^2} \leq \left(\max_i \sum_j (\mathbf{N}_L)_{ij} \right) \|r\|_{\ell^2}.$$

Combining this inequality with (4.5) proves (4.4). ■

We next record the finite-multiplicity perturbation estimate that replaces Lemma 3.1.

Lemma 4.3 (*Isolated spectral-cluster bound*). *Let A be self-adjoint with compact resolvent. Suppose its lowest eigenvalue μ has multiplicity N , let P be the corresponding spectral projection, and suppose*

$$A|_{P^\perp} \geq \mu + g$$

in the quadratic-form sense for some $g > 0$. Let W be bounded and self-adjoint, set $b = \|W\|$, and assume $b \leq g/4$. If $\eta_1 \leq \dots \leq \eta_N$ are the eigenvalues of PWP on $\operatorname{ran} P$, then, for $1 \leq k \leq N$,

$$-\frac{2b^2}{g} \leq \lambda_k(A + W) - (\mu + \eta_k) \leq 0. \quad (4.6)$$

If the underlying Hilbert space has dimension greater than N , then in addition

$$\lambda_{N+1}(A + W) \geq \mu + g - b \geq \mu + \frac{3g}{4}. \quad (4.7)$$

The bound (4.6) holds also when the space is N -dimensional; then $P^\perp = \{0\}$ and the gap condition is vacuous.

Proof. The min-max principle applied to the span of the first k eigenvectors of PWP gives the upper bound in (4.6). For the lower bound, write $\psi = p + q$ with $p = P\psi$ and $q = (I - P)\psi$. Then

$$\begin{aligned} \langle \psi, (A + W)\psi \rangle &\geq \mu \|\psi\|^2 + \langle p, PWPp \rangle + (g - b)\|q\|^2 - 2b\|p\|\|q\| \\ &\geq \mu \|\psi\|^2 + \langle p, PWPp \rangle - \frac{2b^2}{g}\|p\|^2 + \left(\frac{g}{2} - b\right)\|q\|^2. \end{aligned}$$

The second line uses $2b\|p\|\|q\| \leq 2b^2g^{-1}\|p\|^2 + (g/2)\|q\|^2$. Thus, in the form sense, $A + W$ dominates the direct sum of

$$\mu I + PWP - \frac{2b^2}{g}I \quad \text{on } \text{ran } P$$

and $(\mu + g/2 - b)I$ on $P^\perp$. Since $\eta_N \leq b \leq g/4$, we have

$$\eta_N - \frac{2b^2}{g} \leq b - \frac{2b^2}{g} < \frac{g}{2} - b.$$

The strict inequality follows because the difference between the last two quantities is $(g - 2b)^2/(2g) > 0$. Hence the first N eigenvalues of this comparison operator are the N eigenvalues of its P -block. Another application of the min-max principle gives the lower bound in (4.6). Finally, $A + W \geq A - bI$ and the min-max principle give (4.7). $\blacksquare$

Let P_1 be the orthogonal projection in $\mathcal{H}_N$ onto the span of the N vectors whose j th component is ϕ_1 and whose other components vanish. In this basis the compression $P_1 \mathcal{V}_L P_1$ is represented by the real symmetric matrix $T_L = (t_{ij}(L))$, where

$$t_{ii}(L) = 0, \quad t_{ij}(L) = -c_{d,s} \iint_{D \times D} \frac{\phi_1(x)\phi_1(y)}{|L(a_i - a_j) + x - y|^\kappa} dx dy \quad (i \neq j). \quad (4.8)$$

Definition 4.4 (Effective interaction matrix). Define the real symmetric $N \times N$ matrix $M_{\mathbf{a}}$ by

$$(M_{\mathbf{a}})_{ii} = 0, \quad (M_{\mathbf{a}})_{ij} = -c_{d,s} m_1^2 |a_i - a_j|^{-\kappa} \quad (i \neq j). \quad (4.9)$$

Write its ordered eigenvalues as $\theta_1 \leq \dots \leq \theta_N$.

Lemma 4.5 (Compression error). If $L\delta_{\mathbf{a}} \geq 4R$, then

$$\|T_L - L^{-\kappa} M_{\mathbf{a}}\|_{\ell^2 \rightarrow \ell^2} \leq \varepsilon_L, \quad \varepsilon_L = c_{d,s} C_{\kappa,R} m_1^2 \Sigma_{\kappa+2}(\mathbf{a}) L^{-\kappa-2}. \quad (4.10)$$

Here $C_{\kappa,R}$ is given by (3.2).

Proof. Fix $i \neq j$, put $a = a_i - a_j$, and set $F(z) = |La + z|^{-\kappa}$. For $|z| \leq 2R$, the segment from 0 to z stays at distance at least $L|a|/2$ from $-La$. As in the proof of Lemma 3.2, Taylor's theorem gives

$$|F(z) - F(0) - \nabla F(0) \cdot z| \leq C_{\kappa,R} L^{-\kappa-2} |a|^{-\kappa-2}. \quad (4.11)$$

The linear term vanishes after integration because

$$\iint_{D \times D} \phi_1(x) \phi_1(y) (x - y) dx dy = 0.$$

This cancellation is algebraic and does not require central symmetry of D . Using $F(0) = L^{-\kappa}|a|^{-\kappa}$ in (4.8) and integrating (4.11) gives

$$|t_{ij}(L) - L^{-\kappa}(M_{\mathbf{a}})_{ij}| \leq c_{d,s} C_{\kappa,R} m_1^2 L^{-\kappa-2} |a_i - a_j|^{-\kappa-2}.$$

The operator norm of a real symmetric matrix is at most its largest absolute row sum. Taking that row sum proves (4.10). $\blacksquare$

Theorem 4.6 (*Finite multi-well effective interaction matrix*). *Use the fixed one-well data and Definitions 4.1 and 4.4, with γ_L and ε_L from (4.4) and (4.10). If*

$$L\delta_{\mathbf{a}} \geq 4R, \quad \gamma_L \leq \frac{g}{4}, \quad (4.12)$$

then, for $1 \leq k \leq N$,

$$|\lambda_k(\Omega_{\mathbf{a},L}) - (\mu_1 + L^{-\kappa}\theta_k)| \leq \varepsilon_L + \frac{2\gamma_L^2}{g}. \quad (4.13)$$

The upper edge of the cluster and the next eigenvalue satisfy

$$\lambda_N(\Omega_{\mathbf{a},L}) \leq \mu_1 + \gamma_L \leq \mu_1 + \frac{g}{4}, \quad \lambda_{N+1}(\Omega_{\mathbf{a},L}) \geq \mu_1 + g - \gamma_L \geq \mu_1 + \frac{3g}{4}. \quad (4.14)$$

In particular, the cluster gap has the quantitative lower bound

$$\lambda_{N+1}(\Omega_{\mathbf{a},L}) - \lambda_N(\Omega_{\mathbf{a},L}) \geq g - 2\gamma_L \geq \frac{g}{2}. \quad (4.15)$$

Consequently, as $L \rightarrow \infty$, for every $1 \leq k \leq N$,

$$L^{d+2s}(\lambda_k(\Omega_{\mathbf{a},L}) - \mu_1) \longrightarrow \theta_k. \quad (4.16)$$

Proof. Lemma 4.2 identifies the full operator with $\mathcal{A}_0 + \mathcal{V}_L$. The lowest eigenvalue of $\mathcal{A}_0$ is μ_1 , with multiplicity N , its spectral projection is P_1 , and the rest of its spectrum lies in $[\mu_1 + g, \infty)$. Apply Lemma 4.3 with $A = \mathcal{A}_0$, $W = \mathcal{V}_L$, and $b = \|\mathcal{V}_L\| \leq \gamma_L$. If $\tau_1(L) \leq \dots \leq \tau_N(L)$ are the eigenvalues of T_L , this gives

$$-\frac{2\gamma_L^2}{g} \leq \lambda_k(\Omega_{\mathbf{a},L}) - (\mu_1 + \tau_k(L)) \leq 0. \quad (4.17)$$

The finite-dimensional min-max principle and Lemma 4.5 give

$$|\tau_k(L) - L^{-\kappa}\theta_k| \leq \varepsilon_L.$$

The triangle inequality gives (4.13). Using $\tau_N(L) \leq \|T_L\| \leq \|\mathcal{V}_L\| \leq \gamma_L$ in (4.17) gives the upper bound in (4.14); (4.7) gives the lower bound. Their difference proves (4.15). Finally, $L^\kappa \varepsilon_L = O(L^{-2})$ and $L^\kappa \gamma_L^2 = O(L^{-\kappa})$ with $\kappa = d + 2s > 0$, so multiplication of (4.13) by L^κ gives (4.16). $\blacksquare$

Corollary 4.7 (*Symmetry-free two-well asymptotic*). Let $D \subset B_R(0)$ be any nonempty bounded open set, let $a_1 \neq a_2$, and put $r = |a_1 - a_2|$. For $\Omega_L = (D + La_1) \cup (D + La_2)$, as $L \rightarrow \infty$,

$$\begin{aligned}\lambda_1(\Omega_L) &= \mu_1 - c_{d,s} m_1^2 r^{-\kappa} L^{-\kappa} + O(L^{-\kappa-2} + L^{-2\kappa}), \\ \lambda_2(\Omega_L) &= \mu_1 + c_{d,s} m_1^2 r^{-\kappa} L^{-\kappa} + O(L^{-\kappa-2} + L^{-2\kappa}).\end{aligned}$$

In particular,

$$L^{d+2s}(\lambda_2(\Omega_L) - \lambda_1(\Omega_L)) \rightarrow 2c_{d,s} m_1^2 r^{-d-2s}.$$

The constants implicit in the two expansions may depend on d, s, D, R , and r , but are independent of L .

Proof. For $N = 2$, the effective matrix is

$$M_{\mathbf{a}} = \begin{pmatrix} 0 & -c_{d,s} m_1^2 r^{-\kappa} \\ -c_{d,s} m_1^2 r^{-\kappa} & 0 \end{pmatrix},$$

whose ordered eigenvalues are $-c_{d,s} m_1^2 r^{-\kappa}$ and $c_{d,s} m_1^2 r^{-\kappa}$. Theorem 4.6 gives both expansions because $\varepsilon_L = O(L^{-\kappa-2})$ and $\gamma_L^2 = O(L^{-2\kappa})$. Subtracting them and using $\kappa = d + 2s$ gives the limit. $\blacksquare$

Corollary 4.8 (*Two-well splitting without central symmetry*). Let $D \subset B_R(0)$ be any nonempty bounded open set, let e be a unit vector, let $\Omega_L = (D - \frac{L}{2}e) \cup (D + \frac{L}{2}e)$, and let L_0 be given by (3.4). Then every conclusion of Theorem 3.3 holds: for $L \geq L_0$ the bounds (3.5), (3.6), and (3.7) hold and both eigenvalues are simple, and (3.8) holds. For $N = 2$, $a_1 = -e/2$, and $a_2 = e/2$, the error expression $\varepsilon_L + 2\gamma_L^2/g$ in (4.13) equals the right-hand side of each bound (3.5)–(3.6).

Proof. Take $N = 2$, $a_1 = -e/2$, and $a_2 = e/2$, so that $\Omega_{\mathbf{a},L} = \Omega_L$ and $\delta_{\mathbf{a}} = 1$. Then $\Sigma_{\kappa}(\mathbf{a}) = \Sigma_{\kappa+2}(\mathbf{a}) = 1$, so $\gamma_L = \beta_L$ and $\varepsilon_L = c_{d,s} C_{\kappa,R} m_1^2 L^{-\kappa-2}$. Hence

$$\varepsilon_L + \frac{2\gamma_L^2}{g} = c_{d,s} C_{\kappa,R} m_1^2 L^{-\kappa-2} + \frac{2\beta_L^2}{g}.$$

The proof of Theorem 3.3 derives (3.9) from $L \geq L_0$ without using central symmetry; in particular $L \geq 4R$ and $\gamma_L = \beta_L \leq g/4$, so (4.17) and (4.7) apply. By (4.8), the matrix T_L has zero diagonal and off-diagonal entry $-c_{d,s} J_L$, where

$$J_L = \iint_{D \times D} \frac{\phi_1(x)\phi_1(y)}{|Le - x + y|^\kappa} dx dy,$$

so its ordered eigenvalues are $\mp c_{d,s} J_L$. Lemma 4.5 gives $|J_L - m_1^2 L^{-\kappa}| \leq C_{\kappa,R} m_1^2 L^{-\kappa-2}$, and (4.17) then gives (3.5) and (3.6).

Moreover, $c_{d,s} J_L \leq \|T_L\| \leq \|\mathcal{V}_L\| \leq \beta_L$. Hence $\lambda_2(\Omega_L) \leq \mu_1 + g/4$. Also, by (4.7), $\lambda_3(\Omega_L) \geq \mu_1 + 3g/4$. This proves (3.7). Since $\phi_1 \geq 0$ and $|Le - x + y| \leq L + 2R$ on $D \times D$, we have $J_L \geq m_1^2 (L + 2R)^{-\kappa}$, and (4.17) gives

$$\lambda_2(\Omega_L) - \lambda_1(\Omega_L) \geq 2c_{d,s} J_L - \frac{2\beta_L^2}{g} \geq 2c_{d,s} m_1^2 (L + 2R)^{-\kappa} - \frac{2\beta_L^2}{g} > 0$$

by the second inequality in (3.9). Both eigenvalues are therefore simple. Subtracting (3.5) from (3.6) and multiplying by L^κ proves (3.8). When $D = -D$, evenness of ϕ_1 and the change of variables $y \mapsto -y$ give $J_L = I_L$, so the leading terms coincide with the reflected branches of Theorem 3.3. ■

Corollary 4.9 (*Collective ground state*). *For every multi-well configuration, $\theta_1 < 0$ is simple and admits an eigenvector with strictly positive entries, while $\theta_N > 0$. For every L satisfying the separation condition $L\delta_{\mathbf{a}} > 2R$ in Definition 4.1, $\lambda_1(\Omega_{\mathbf{a},L})$ is simple. Moreover, for all sufficiently large L , $\lambda_1(\Omega_{\mathbf{a},L}) < \mu_1 < \lambda_N(\Omega_{\mathbf{a},L})$.*

Proof. The off-diagonal entries of $M_{\mathbf{a}}$ are strictly negative, and its Rayleigh quotient at $(1, \dots, 1)$ is negative, so $\theta_1 < 0$. Replacing a Rayleigh minimizer z by its componentwise absolute value cannot increase the quotient; equality forces its nonzero components to have the same sign, and the eigenvalue equation rules out a zero component. Thus the components of a ground-state eigenvector are either all strictly positive or all strictly negative. If the ground eigenspace had dimension at least two, it would contain a nonzero vector with a zero component, a contradiction. Hence θ_1 is simple. Finally, $\text{tr } M_{\mathbf{a}} = 0$ gives $\theta_N > 0$. Put $r_L = \varepsilon_L + 2\gamma_L^2/g$. Theorem 4.6 gives $L^\kappa r_L \rightarrow 0$. Simplicity of $\lambda_1(\Omega_{\mathbf{a},L})$ for every admissible L follows from Lemma 2.2, applied to the nonempty bounded open set $\Omega_{\mathbf{a},L}$. Choose L so large that

$$L^\kappa r_L < \min\{-\theta_1, \theta_N\}.$$

Then (4.13) gives $\lambda_1 < \mu_1 < \lambda_N$, which proves the remaining assertions. ■

Corollary 4.10 (*Regular-simplex cluster*). *Suppose the sites satisfy $|a_i - a_j| = r$ for all $i \neq j$, as for the vertices of a regular simplex. Then $N \leq d + 1$. Put*

$$w = c_{d,s} m_1^2 r^{-\kappa}.$$

Then $\theta_1 = -(N-1)w$ and $\theta_2 = \dots = \theta_N = w$. Thus one collective level is lowered by $(N-1)wL^{-\kappa}$, while the remaining $N-1$ levels are raised by $wL^{-\kappa}$, up to the remainder in (4.13).

Proof. After translating a_N to the origin, the $N-1$ vectors $a_i - a_N$ have Gram matrix with diagonal entries r^2 and off-diagonal entries $r^2/2$. This matrix is positive definite, so $N-1 \leq d$. Here $M_{\mathbf{a}} = -w(J - I)$, where J is the all-ones matrix. The vector $(1, \dots, 1)$ has eigenvalue $-(N-1)w$, and every vector in its orthogonal complement is an eigenvector with eigenvalue w . ■

5. Numerical illustrations on intervals

We illustrate the fixed-mesh cluster asymptotic of Corollary 5.2 using conforming Galerkin matrices assembled from exact cell integrals. The mesh is held fixed as L increases; no discretization-error estimate as $h \rightarrow 0$ is asserted. Take $d = 1$, $0 < s < 1/2$, and consider finite unions of unit intervals. For high-index eigenvalue asymptotics and eigenfunction estimates on a single interval, see [Kwa12, Theorem 1 and Propositions 1–2]; sharper eigenvalue

asymptotics and an eigenfunction bound uniform in both the index and the fractional order are proved in [Zha26, Theorems 1 and 2]. Both sources parameterize the operator order as $\alpha = 2s$.

Partition each component into cells of length h . The cell indicators have finite form energy by Proposition 5.1, so they span a conforming finite-dimensional subspace of the form domain.

Proposition 5.1 (*Exact cell-integral matrix*). Assume $0 < s < 1/2$. Let $I_1, \dots, I_M$ be pairwise disjoint cells of common length h , with centers x_α , and set $G = \bigcup_{\alpha=1}^M I_\alpha$, $r_{\alpha\beta} = |x_\alpha - x_\beta|$, and $q = 1 - 2s$. In the $L^2(G)$ -orthonormal basis $e_\alpha = h^{-1/2} \mathbf{1}_{I_\alpha}$, the Galerkin matrix H_h of the zero-exterior form $\mathcal{Q}_G$ has entries

$$(H_h)_{\alpha\alpha} = \frac{c_{1,s}}{h} \frac{2h^q}{2sq}, \quad (5.1)$$

$$(H_h)_{\alpha\beta} = -\frac{c_{1,s}}{h} \frac{2r_{\alpha\beta}^q - (r_{\alpha\beta} + h)^q - (r_{\alpha\beta} - h)^q}{2sq}, \quad \alpha \neq \beta. \quad (5.2)$$

Proof. For two disjoint cells with center distance $r \geq h$, two direct integrations give

$$\iint_{I_\alpha \times I_\beta} |x - y|^{-1-2s} dx dy = \frac{2r^q - (r + h)^q - (r - h)^q}{2sq}.$$

Multiplication by $-c_{1,s}/h$ gives (5.2). Similarly, the diagonal entry in (5.1) is $c_{1,s}/h$ times

$$\int_{I_\alpha} \int_{\mathbb{R} \setminus I_\alpha} |x - y|^{-1-2s} dy dx = \frac{2h^q}{2sq}. \quad \blacksquare$$

Corollary 5.2 (*Fixed-mesh cluster asymptotic*). Let $D = (-1/2, 1/2)$, and fix a uniform partition of D . Let $\mu_{1,h}$ and $\phi_{1,h}$ be the lowest eigenvalue and a positive, L^2 -normalized piecewise-constant eigenfunction of the one-well Galerkin matrix. Put $m_{1,h} = \int_D \phi_{1,h}$. Translate this same partition to each component $D + La_j$, and write the ordered eigenvalues of the resulting Galerkin matrix on $\bigcup_{j=1}^N (D + La_j)$ as $\lambda_{k,h}(L)$. Then, as $L \rightarrow \infty$, for fixed distinct $a_1, \dots, a_N$,

$$\lambda_{k,h}(L) = \mu_{1,h} + L^{-1-2s} \theta_{k,h} + O_{h,\mathbf{a},s}(L^{-3-2s} + L^{-2-4s}), \quad 1 \leq k \leq N, \quad (5.3)$$

where $\theta_{1,h} \leq \dots \leq \theta_{N,h}$ are the eigenvalues of the matrix with zero diagonal and off-diagonal entries $-c_{1,s} m_{1,h}^2 |a_i - a_j|^{-1-2s}$.

Proof. All off-diagonal entries of the one-well matrix in (5.2) are negative. The Rayleigh-quotient argument in the proof of Corollary 4.9 therefore gives a simple lowest eigenvalue and a strictly positive eigenvector. On the direct sum of N fixed Galerkin spaces, the block expansion used in Lemma 4.2 is exact, its off-diagonal norm is $O_{h,\mathbf{a},s}(L^{-1-2s})$, and the complement of the N copied ground states is separated by the positive discrete one-well gap. The Taylor argument in Lemma 4.5, now in Euclidean operator norm, gives an $O_{h,\mathbf{a},s}(L^{-3-2s})$ compression error. If the partition has a single cell, the Galerkin space is spanned by the N copied ground states and only (4.6) is needed. Applying Lemma 4.3 to this finite matrix gives the additional $O_{h,\mathbf{a},s}(L^{-2-4s})$ term and proves (5.3). $\blacksquare$

For the computation we take $s = 1/4$, $D = (-1/2, 1/2)$, and the standard normalization

$$c_{1,s} = \frac{4^s s \Gamma(1/2 + s)}{\sqrt{\pi} \Gamma(1 - s)}.$$

Table 1 records the one-well quantities, including the dependence on mesh size of $m_{1,h}$, which enters the interaction coefficient.

cells	$\mu_{1,h}$	$\mu_{2,h} - \mu_{1,h}$	$m_{1,h}$
60	1.379455	0.910554	0.962839
120	1.375369	0.899888	0.963351
180	1.374129	0.896965	0.963602

Table 1. One-well Galerkin data for $D = (-1/2, 1/2)$, $s = 1/4$, and the indicated uniform meshes.

On the 180-cell-per-component mesh, the two-well component centers are $(-L/2, L/2)$, while the three-well centers are $(-L, 0, L)$. For the corresponding fixed sites, the effective matrices on this mesh have eigenvalues

$$\begin{aligned} \mathbf{a} = (-1/2, 1/2) : & \quad (-0.18521477, 0.18521477), \\ \mathbf{a} = (-1, 0, 1) : & \quad (-0.29671332, 0.06548331, 0.23123001). \end{aligned}$$

Table 2 lists the scaled shifts computed from the Galerkin eigenvalues. For each configuration, the last column is

$$E_h(L) = \max_{1 \leq k \leq N} |L^{3/2}(\lambda_{k,h}(L) - \mu_{1,h}) - \theta_{k,h}|. \quad (5.4)$$

These errors are computed from the full-precision values before the shifts and effective spectra are rounded for display.

At the selected separations, the scaled errors decrease monotonically; Corollary 5.2 proves that they tend to zero at fixed mesh. Here $g_h = \mu_{2,h} - \mu_{1,h}$ and $R = 1/2$. At $L = 3$, the bound γ_L in (4.4) is 0.10858 for two wells and 0.21716 for three wells, while $g_h/4 = 0.22424$. Since γ_L decreases with L and $L\delta_{\mathbf{a}} \geq 4R$ already at $L = 3$, the finite-dimensional cluster estimate used in the proof applies at every computed separation. The calculation uses the exact entries (5.1)–(5.2); no singular quadrature or fitted coefficient enters the tables.

<i>Two wells</i>				
L	$L^{3/2}(\lambda_{1,h} - \mu_{1,h})$	$L^{3/2}(\lambda_{2,h} - \mu_{1,h})$	$E_h(L)$	
3	-0.190740	0.189832	5.52×10^{-3}	
6	-0.186544	0.186335	1.33×10^{-3}	
12	-0.185551	0.185486	3.36×10^{-4}	
24	-0.185302	0.185279	8.68×10^{-5}	

<i>Three wells</i>				
L	$L^{3/2}(\lambda_{1,h} - \mu_{1,h})$	$L^{3/2}(\lambda_{2,h} - \mu_{1,h})$	$L^{3/2}(\lambda_{3,h} - \mu_{1,h})$	$E_h(L)$
3	-0.304984	0.065562	0.237544	8.27×10^{-3}
6	-0.298738	0.065554	0.232747	2.03×10^{-3}
12	-0.297236	0.065504	0.231595	5.22×10^{-4}
24	-0.296852	0.065488	0.231316	1.38×10^{-4}

Table 2. Scaled cluster shifts for $s = 1/4$ and $D = (-1/2, 1/2)$ on the 180-cell-per-component mesh, with two wells centered at $(-L/2, L/2)$ or three wells centered at $(-L, 0, L)$. The maximum scaled-shift error $E_h(L)$ is defined in (5.4).

6. Formal verification

The theorems, the corollaries, and Proposition 5.1, together with the one-well facts of Section 2 that they use, are formalized in Lean 4 (version 4.33.0) with the Mathlib library. The formalization is in the directory `lean` of the repository listed under Code availability, at <https://github.com/skypher/laplace-tunneling/tree/paper-2026-09-30-r2/lean>. A continuous-integration job rebuilds it on every change. The job fails if any proof is incomplete, if the project declares an axiom, or if a result in Table 3 depends on axioms other than the three standard ones of Lean (propositional extensionality, choice, and quotient soundness).

The operator A_G enters through its closed form. The space $\tilde{H}^s(G)$ is a subspace of $L^2(\mathbb{R}^d)$, (2.1) is a quadratic form on it, and $\lambda_k(G)$ is defined by the Courant–Fischer min–max principle. For every nonempty bounded open set G , the formalization proves that these numbers are exactly the eigenvalues of the weak problem $\mathcal{Q}_G(u, v) = \lambda \langle u, v \rangle$ for all $v \in \tilde{H}^s(G)$, repeated according to multiplicity. It also proves Lemmas 2.1 and 2.2, including the uniqueness of the nonnegative normalized ground state ϕ_1 of (2.2), and the gap inequality on the orthogonal complement of ϕ_1 . Strict positivity of ϕ_1 is not formalized; it is cited only for Lipschitz wells in Section 2 and is not used.

The formal statements of Sections 2–4 hold for every nonempty bounded open set $D \subset B_R(0)$, all $d \geq 1$ and $0 < s < 1$, and every positive constant in place of $c_{d,s}$; those of Section 5 keep the assumptions $d = 1$ and $0 < s < 1/2$ stated there. Lemma 4.2 is formalized in the sense of closed forms, which is how the theorems use it. The formal proof of Theorem 3.3 follows Corollary 4.8; Lemmas 2.3, 3.1, and 3.2, which enter only the symmetric proof, are not formalized as stated. The numerical tables are not part of the formalization.

Result	Lean declaration
Lemma 2.1	<code>totallyBounded_energySublevel</code>
Lemma 2.2	<code>exists_positiveGroundState, eigenvalue_zero_lt_one</code>
Weak eigenvalues	<code>finrank_weakEigenspace</code>
Theorem 3.3	<code>twoWell_main_centrallySymmetric</code>
Corollary 3.4	<code>distantBall_rate</code>
Lemma 4.2	<code>WellGeometry.formQ_eq, WellGeometry.abs_interaction_le</code>
Lemma 4.3	<code>MinMax.cluster_levels, MinMax.cluster_levels_bounds</code>
Lemma 4.5	<code>compression_norm_le_euclid</code>
Theorem 4.6	<code>multiWell_main</code>
Corollary 4.7	<code>symmetryFree_twoWell</code>
Corollary 4.8	<code>twoWell_main</code>
Corollary 4.9	<code>collectiveGround_matrix, collectiveGround_spectrum</code>
Corollary 4.10	<code>simplex_cluster</code>
Proposition 5.1	<code>cellMatrix_diag, cellMatrix_offdiag</code>
Corollary 5.2	<code>fixedMesh_cluster</code>

Table 3. Formal counterparts of the results, in the namespace `Tunneling`.

AI disclosure

The author used OpenAI’s GPT-5.6 Sol model (`gpt-5.6-sol`) to explore and test arguments, audit citations and calculations, prepare reproducibility and release-checking code, and revise the manuscript. The Lean formalization of Section 6 was developed with Anthropic’s Claude Opus 5.5 model (`claude-opus-5-5`) and OpenAI’s GPT-6 Luna model (`gpt-6-luna`); its correctness rests on the Lean proof checker. The author checked the final arguments and takes responsibility for all mathematical claims, citations, computations, and wording.

Code availability

The script `check_asymptotics.py` reproduces Tables 1 and 2. The ancillary files also include the pinned dependencies in `requirements-numeric.txt` and the reference output in `asymptotics_180.txt`. The calculation was reproduced using Python 3.12.3, NumPy 1.26.4, and SciPy 1.11.4. These files are archived with the manuscript and are also available, with the manuscript source, in the GitHub repository <https://github.com/skypher/laplace-tunneling> (tag `paper-2026-09-30-r2`). The Lean formalization of Section 6 and its build configuration are in the directory `lean` of the same repository and tag.

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